



LANGUAGE MASTER

HANDOUT

The Exciting Future of 3D Printing
Academic Reading Circles



KEY VOCABULARY

1. Voxel (noun)

Definition: A volume pixel - the smallest 3D unit that a printer can produce (like a 3D version of a pixel)

Example from text: "adding material layer by layer or voxel by voxel (voxel=volume pixel)"

2. Biocompatibility (noun)

Definition: Compatibility with biological systems; being safe for living cells

Example from text: "These bio-inks enable the creation of 3D structures with sub-micron resolution, low toxicity, and biocompatibility"

3. Photoresin (noun)

Definition: A liquid that consists of monomers and photoinitiator molecules that solidify when exposed to light

Example from text: "a photosensitive resin is hardened, a process typically referred to as 'curing'"

4. Scalability (noun)

Definition: How well a process can be adapted for mass production or large-scale manufacturing

Example from text: "3DMM2O researchers aim to improve spatial resolution, enhance manufacturing scalability"

5. Nanoscale (noun/adjective)

Definition: Objects with sub-micrometre features, typically ranging from 1 to 1000 nanometres in size

Example from text: "Moving 3D printing towards the nanoscale by using technologically advanced light-based microprinters"

6. Thermoplastic (noun/adjective)

Definition: A material that softens when heated and solidifies when cooled, and can be reshaped multiple times

Example from text: "FDM works by heating a thermoplastic material, typically in the form of a filament"



TOPICAL VOCABULARY

Category 1: PRINTING TECHNOLOGY

Word/Phrase	Word Family
3D printing	print (v), printer (n), printable (adj), printing (n)

polymerization	polymer (n), polymerize (v), polymeric (adj)
light-based	light (n), illuminate (v), illumination (n)
extrusion	extrude (v), extruded (adj), extruder (n)
photosensitive	photo (n), sensitive (adj), sensitivity (n)

Category 2: MATERIALS & PROCESSES

Word/Phrase	Word Family
thermoplastic	plastic (n/adj), plasticity (n)
filament	fiber (n), fibrous (adj)
solidify	solid (n/adj), solidification (n)
resin	resinous (adj)
curing	cure (v), cured (adj)

Category 3: SCIENCE & RESEARCH

Word/Phrase	Word Family
biocompatibility	compatible (adj), compatibility (n), incompatible (adj)
nanoscale	nano (prefix), scale (n), microscale (n)
spheroid	sphere (n), spherical (adj)
sustainable	sustain (v), sustainability (n), unsustainable (adj)
microstructure	structure (n), structural (adj), macro (prefix)



ACADEMIC LANGUAGE

Passive Structures:

- ▶ "3D printing has been a fixture in science fiction..."
- ▶ "These bonds are exceptionally strong"
- ▶ "the printed objects exhibit exceptional properties"
- ▶ "researchers are exploring microalgae oil"

Nominalisations:

- ▶ **polymerization** (from: polymerize)
- ▶ **democratization** (from: democratize)
- ▶ **confinement** (from: confine)
- ▶ **adaptability** (from: adapt)

Formal Connectors:

- ▶ **"In contrast"** - showing difference
- ▶ **"Consequently"** - showing result
- ▶ **"Additionally"** - adding information
- ▶ **"However"** - showing contrast



TONAL LANGUAGE

Positive/Enthusiastic words:

- "exciting future"
- "transformative technology"
- "ground-breaking applications"
- "invaluable tool"

Scientific/Neutral words:

- "typically referred to as"
- "researchers have developed"
- "significantly less waste"



LANGUAGE EXERCISES

Exercise 1: MATCHING - Connect the word with its definition

WORD	DEFINITION
1. Voxel	A. A material that softens when heated and hardens when cooled
2. Biocompatibility	B. The smallest 3D unit a printer can make
3. Photoresin	C. How well something works with living cells
4. Scalability	D. Objects sized from 1 to 1000 nanometres

5. Nanoscale

E. A liquid that hardens when exposed to light

6. Thermoplastic

F. How well a process can be used for mass production

Exercise 2: GAP-FILL

Complete the sentences with words from the box:

layer

curing

resolution

sustainable

microstructure

extrusion

1. ▶ The printer builds objects _____ by _____.
2. ▶ _____ refers to how small a detail the printer can make.
3. ▶ The _____ process uses light to harden the resin.
4. ▶ Using microalgae makes 3D printing more _____.
5. ▶ _____-based printers push heated plastic through a nozzle.
6. ▶ Scientists can now examine the _____ of printed objects at the micrometre level.



ANSWERS

Exercise 1: Matching

1-B, 2-C, 3-E, 4-F, 5-D, 6-A

Exercise 2: Gap-fill

1. ▶ layer by layer
2. ▶ Resolution
3. ▶ curing
4. ▶ sustainable
5. ▶ Extrusion
6. ▶ microstructure